16

Wave Motion

CHAPTER OUTLINE

16.1 Propagation of a Disturbance

16.2 Analysis Model: Traveling Wave

16.3 The Speed of Transverse Waves on Strings

16.4 Reflection and Transmission

16.5 Rate of Energy Transfer by Sinusoidal Waves on Strings

16.6 The Linear Wave Equation

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ16.1 (i) Answer (a). As the wave passes from the massive string to the less massive string, the wave speed will increase according to $v = \sqrt{\frac{T}{\mu}}$.

(ii) Answer (c). The frequency will remain unchanged. The rate at which crests come up to the boundary is the same rate at which they leave the boundary.

(iii) Answer (a). Since $v = f\lambda$, the wavelength must increase.

OQ16.2 (i) Answer (a). Higher tension makes wave speed higher.

(ii) Answer (b). Greater linear density makes the wave move more slowly.

OQ16.3 (i) The ranking is (c) = (d) > (e) > (b) > (a). Look at the coefficients of the sine and cosine functions: (a) 4, (b) 6, (c) 8, (d) 8, (e) 7.

(ii) The ranking is (c) > (a) = (b) > (d) > (e). Look at the coefficients of x. Each is the wave number, $2\pi/\lambda$, so the smallest k goes with

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the largest wavelength.

(iii) The ranking is (e) > (d) > (a) = (b) = (c). Look at the coefficients of t. The absolute value of each is the angular frequency $\omega = 2\pi f$.

(iv) The ranking is (a) = (b) = (c) > (d) > (e). Period is the reciprocal of frequency, so the ranking is the reverse of that in part (iii).

(v) The ranking is (c) > (a) = (b) = (d) > (e). From $v = f\lambda = \omega / k,$ we compute the absolute value of the ratio of the coefficient of t to the coefficient of x in each case: (a) 5, (b) 5, (c) 7.5, (d) 5, (e) 4.

OQ16.4 Answer (b). From $v = \sqrt{\frac{T}{\mu}},$ we must increase the tension by a factor of 4 to make v double.

OQ16.5 Answer (b). Wave speed is inversely proportional to the square root of linear density.

OQ16.6 Answer (b). Not all waves are sinusoidal. A sinusoidal wave is a wave of a single frequency. In general, a wave can be a superposition of many sinusoidal waves.

OQ16.7 (a) through (d): Yes to all. The maximum element speed and the wave speed are related by $v_{y,\max} = \omega A = 2\pi fA = 2\pi vA / \lambda.$ Thus the amplitude or the wavelength of the wave can be adjusted to make either $v_{y,\max}$ or v larger.

OQ16.8 Answer (c). The power carried by a wave is proportional to its frequency, wave speed, and the square of its amplitude. If the frequency does not change, the amplitude is increased by a factor of $\sqrt{2}.$ The wave speed does not change.

OQ16.9 Answer (c). The distance between two successive peaks is the wavelength: λ = 2 m, and the frequency is 4 Hz. The frequency, wavelength, and speed of a wave are related by the equation $f\lambda = v$.

ANSWERS TO CONCEPTUAL QUESTIONS

CQ16.1 Longitudinal waves depend on the compressibility of the fluid for their propagation. Transverse waves require a restoring force in response to shear strain. Fluids do not have the underlying structure to supply such a force. A fluid cannot support static shear. A viscous fluid can temporarily be put under shear, but the higher its viscosity the more quickly it converts kinetic energy into internal energy. A local vibration imposed on it is strongly damped, and not a source of wave propagation.

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CQ16.2 The type of wave you generate depends upon the direction of the disturbance (vibration) you generate and the direction of its travel (propagation).

(a) To use a spring (or slinky) to create a longitudinal wave, pull a few coils back and release.

(b) For a transverse wave, jostle the end coil side to side.

CQ16.3 It depends on from what the wave reflects. If reflecting from a less dense string, the reflected part of the wave will be right side up. A wave inverts when it reflects off a medium in which the wave speed is smaller.

CQ16.4 The speed of a wave on a "massless" string would be infinite!

CQ16.5 Since the frequency is 3 cycles per second, the period is 1/3 second = 333 ms.

CQ16.6 (a) and (b) Each element of the rope must support the weight of the rope below it. The tension increases with height. (It increases linearly, if the rope does not stretch.) Then the wave speed $v = \sqrt{\frac{T}{\mu}}$ increases with height.

CQ16.7 As the pulse moves down the string, the elements of the string itself move side to side. Since the medium—here, the string—moves perpendicular to the direction of wave propagation, the wave is transverse by definition.

CQ16.8 No. The vertical speed of an element will be the same on any string because it depends only on frequency and amplitude:

$$v_{y,\max} = \omega A = 2\pi fA$$

The elements of strings with different wave speeds will have the same maximum vertical speed.

CQ16.9 (a) Let $\Delta t = t_s - t_p$ represent the difference in arrival times of the two waves at a station at distance $d = v_s t_s = v_p t_p$ from the focus. Then $d = \Delta t \left(\frac{1}{v_s} - \frac{1}{v_p} \right)^{-1}$.

(b) Knowing the distance from the first station places the focus on a sphere around it. A measurement from a second station limits it to another sphere, which intersects with the first in a circle. Data from a third non-collinear station will generally limit the possibilities to a point.

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SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 16.1 Propagation of a Disturbance

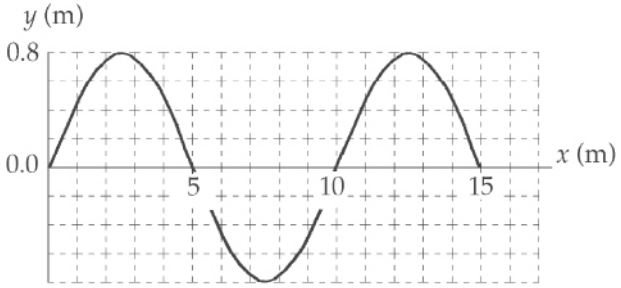
P16.1 The distance the waves have traveled is $d = (7.80\ \text{km/s})t = (4.50\ \text{km/s})(t + 17.3\ \text{s})$, where t is the travel time for the faster wave.

Then, $$(7.80 - 4.50)(\text{km/s})t = (4.50\ \text{km/s})(17.3\ \text{s})$$

or $$t = \frac{(4.50\ \text{km/s})(17.3\ \text{s})}{(7.80 - 4.50)\ \text{km/s}} = 23.6\ \text{s}$$

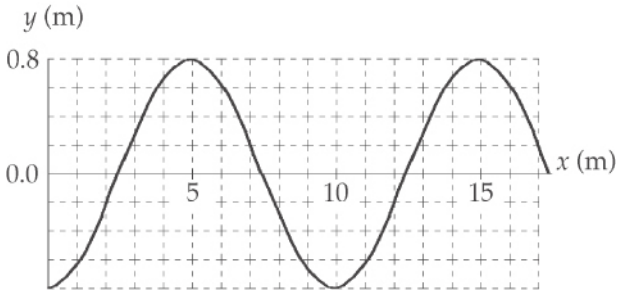
and the distance is $d = (7.80\ \text{km/s})(23.6\ \text{s}) =$ **184 km**

P16.2 (a) ANS. FIG. P16.2(a) shows the sketch of $y(x,t)$ at $t = 0$.



ANS. FIG. P16.2(a)

(b) ANS. FIG. P16.2(b) shows the sketch of $y(x,t)$ at $t = 2.00$ s.



ANS. FIG. P16.2(b)

(c) The graph in ANS. FIG. P16.2(b) has the same amplitude and wavelength as the graph in ANS. FIG. P16.2(a). It differs just by being shifted toward larger x by 2.40 m.

(d) The wave has travelled $d = vt = 2.40$ m to the right.

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P16.3 We obtain a function of the same shape by writing

$$y(x,t)=\frac{6}{\left[\left(x-x_0\right)^2+3\right]}$$

where the center of the pulse is at $x_0 = 4.50t$. Thus, we have

$$\boxed{y=\frac{6}{\left[(x-4.50t)^2+3\right]}}$$

Note that for y to stay constant as t increases, x must increase by $4.50t$, as it should to describe the wave moving at 4.50 m/s.

P16.4 (a) The $\boxed{\text{longitudinal P wave}}$ travels a shorter distance and is moving faster, so it will arrive at point B first.

(b) The P wave that travels through the Earth must travel

a distance of $2R\sin 30.0^\circ = 2\left(6.37\times10^6\ \text{m}\right)\sin 30.0^\circ = 6.37\times10^6\ \text{m}$

at a speed of 7 800 m/s.

Therefore, it takes $\Delta t_P = \dfrac{6.37\times10^6\ \text{m}}{7\,800\ \text{m/s}} \simeq 817\ \text{s}.$

The Rayleigh wave that travels along the Earth's surface must travel a distance of

$$s = R\theta = R\left(\frac{\pi}{3}\ \text{rad}\right) = 6.67\times10^6\ \text{m}$$

at a speed of 4 500 m/s.

Therefore, it takes $\Delta t_S = \dfrac{6.67\times10^6\ \text{m}}{4\ 500\ \text{m/s}} \simeq 1\ 482\ \text{s}.$

The time difference is $\Delta T = \Delta t_S - \Delta t_P = \boxed{666\ \text{s}} = 11.1\ \text{min}.$

Section 16.2 Analysis Model: Traveling Wave

P16.5 Compare the specific equation to the general form:

$$y = (0.020\ 0\ \text{m})\sin(2.11x - 3.62t) = y = A\sin(kx - \omega t + \phi)$$

(a) $A = \boxed{2.00\ \text{cm}}$

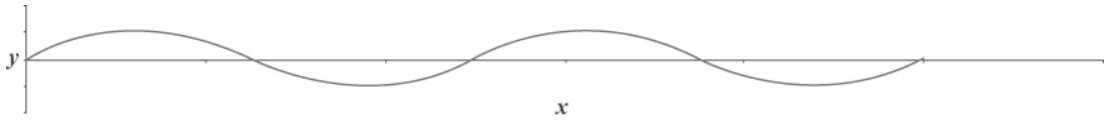
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(b) $k = 2.11 \text{ rad/m} \rightarrow \lambda = \dfrac{2\pi}{k} = \boxed{2.98 \text{ m}}$

(c) $\omega = 3.62 \text{ rad/s} \rightarrow f = \dfrac{\omega}{2\pi} = \boxed{0.576 \text{ Hz}}$

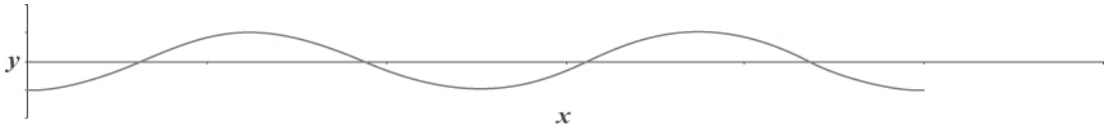
(d) $v = f\lambda = \dfrac{\omega}{2\pi}\dfrac{2\pi}{k} = \dfrac{3.62}{2.11} = \boxed{1.72 \text{ m/s}}$

P16.6 (a) ANS. FIG. P16.6(a) shows the snapshot of a wave on a string.



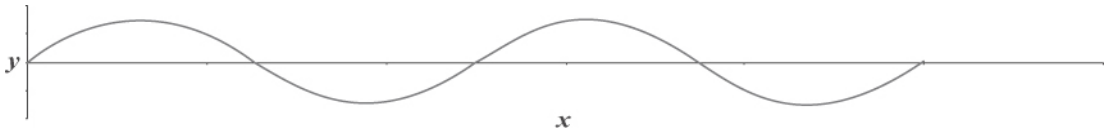
ANS. FIG. P16.6(a)

(b) ANS. FIG. P16.6(b) shows the wave from part (a) one-quarter period later



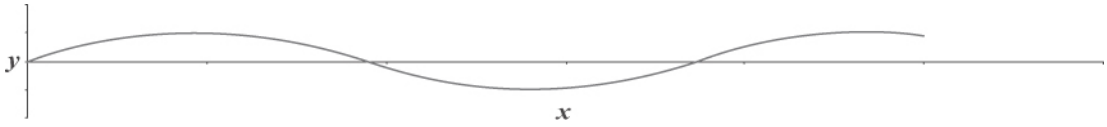
ANS. FIG. P16.6(b)

(c) ANS. FIG. P16.6(c) shows a wave with an amplitude 1.5 times larger than the wave in part (a).



ANS. FIG. P16.6(c)

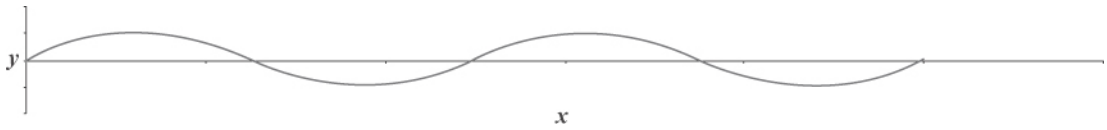
(d) ANS. FIG. P16.6(d) shows a wave with wavelength 1.5 times larger than the wave in part (a).



ANS. FIG. P16.6 (d)

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(e) ANS. FIG. P16.6(e) shows a wave with frequency 1.5 times larger than the wave in part (a): The wave appears the same as in ANS. FIG. P16.6(a) because this is a snapshot of a given moment.



ANS. FIG. P16.6(e)

P16.7 The frequency of the wave is

$$f = \frac{40.0 \text{ vibrations}}{30.0 \text{ s}} = \frac{4}{3} \text{ Hz}$$

as the wave travels 425 cm in 10.0 s, its speed is

$$v = \frac{425 \text{ cm}}{10.0 \text{ s}} = 42.5 \text{ cm/s}$$

and its wavelength is therefore

$$\lambda = \frac{v}{f} = \frac{42.5 \text{ cm/s}}{1.33 \text{ Hz}} = 31.9 \text{ cm} = \boxed{0.319 \text{ m}}$$

P16.8 Using data from the observations, we have $\lambda = 1.20$ m and

$$f = \frac{8.00 \text{ crests}}{12.0 \text{ s}} = \frac{8.00 \text{ cycles}}{12.0 \text{ s}} = \frac{8.00}{12.0} \text{ Hz}$$

Therefore, $v = \lambda f = (1.20 \text{ m})\left(\frac{8.00}{12.0} \text{ Hz}\right) = \boxed{0.800 \text{ m/s}}$.

P16.9 (a) We note that $\sin\theta = -\sin(-\theta) = \sin(-\theta + \pi)$, so the given wave function can be written as

$$y(x,t) = (0.350)\sin(-10\pi t + 3\pi x + \pi - \pi/4)$$

Comparing, $10\pi t - 3\pi x + \pi/4 = kx - \omega t + \phi$. For constant phase, x must increase as t increases, so the wave travels in the positive x direction. Comparing the specific form to the general form, we find that

$$v = \frac{\omega}{k} = \frac{10\pi}{3\pi} = 3.33 \text{ m/s}.$$

Therefore, the velocity is $\boxed{\left(3.33\hat{\mathbf{i}}\right) \text{ m/s}}$.

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(b) Substituting $t = 0$ and $x = 0.100$ m, we have

$$y(0.100\ 0) = (0.350\ \text{m}) \sin\left(-0.300\pi + \frac{\pi}{4}\right) = -0.054\ 8\ \text{m}$$

$$= \boxed{-5.48\ \text{cm}}$$

(c) $k = \dfrac{2\pi}{\lambda} = 3\pi$: $\lambda = \boxed{0.667\ \text{m}}$ $\omega = 2\pi f = 10\pi$: $f = \boxed{5.00\ \text{Hz}}$

(d) $v_y = \dfrac{\partial y}{\partial t} = (0.350)(10\pi)\cos\left(10\pi t - 3\pi x + \dfrac{\pi}{4}\right)$

$$v_{y,\,\max} = (10\pi)(0.350) = \boxed{11.0\ \text{m/s}}$$

P16.10 The speed of waves along this wire is

$$v = f\lambda = (4.00\ \text{Hz})(60.0\ \text{cm}) = 240\ \text{cm/s} = \boxed{2.40\ \text{m/s}}$$

P16.11 (a) $\omega = 2\pi f = 2\pi\left(5.00\ \text{s}^{-1}\right) = \boxed{31.4\ \text{rad/s}}$

(b) $\lambda = \dfrac{v}{f} = \dfrac{20.0\ \text{m/s}}{5.00\ \text{s}^{-1}} = 4.00\ \text{m}$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{4.00\ \text{m}} = \boxed{1.57\ \text{rad/m}}$$

(c) In $y = A\sin\left(kx - \omega t + \phi\right)$ we take $A = 12.0$ cm. At $x = 0$ and $t = 0$ we have $y = (12.0\ \text{cm})\sin\phi$. To make this fit $y = 0$, we take $\phi = 0$. Then

$\boxed{y = 0.120\sin(1.57x - 31.4t)\text{, where } x \text{ and } y \text{ are in meters and } t \text{ is in seconds}}$

(d) The transverse velocity is $\dfrac{\partial y}{\partial t} = -A\omega\cos(kx - \omega t)$.

Its maximum magnitude is

$$A\omega = (12.0\ \text{cm})\left(31.4\ \text{rad/s}\right) = \boxed{3.77\ \text{m/s}}$$

(e) $a_y = \dfrac{\partial v_y}{\partial t} = \dfrac{\partial}{\partial t}\left[-A\omega\cos(kx - \omega t)\right] = -A\omega^2\sin(kx - \omega t)$

The maximum value is $A\omega^2 = (0.120\ \text{m})\left(31.4\ \text{s}^{-1}\right)^2 = \boxed{118\ \text{m/s}^2}$

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P16.12 At time t, the motion at point A, where $x = 0$, is

$$y_A = (1.50\text{ cm})\cos(-50.3t)$$

At point B, the motion is

$$y_B = (15.0\text{ cm})\cos(15.7x_B - 50.3t) = (15.0\text{ cm})\cos\left(-50.3t \pm \frac{\pi}{3}\right)$$

which implies

$$15.7x_B = (15.7\text{ m}^{-1})x_B = \pm\frac{\pi}{3}$$

or $$x_B = -0.066\ 7\text{ m} = \boxed{\pm 6.67\text{ cm}}$$

P16.13 (a) $$f = \frac{v}{\lambda} = \frac{(1.00\text{ m/s})}{2.00\text{ m}} = \boxed{0.500\text{ Hz}}$$

(b) $$\omega = 2\pi f = 2\pi(0.500/\text{s}) = \pi/\text{s} = \boxed{3.14\text{ rad/s}}$$

(c) $$k = \frac{2\pi}{\lambda} = \frac{2\pi}{2.00\text{ m}} = \pi/\text{m} = \boxed{3.14\text{ rad/m}}$$

(d) $y = A\sin(kx - \omega t + \phi)$ becomes

$$y = \boxed{0.100\sin(\pi x - \pi t)}$$

(e) For $x = 0$ the wave function requires

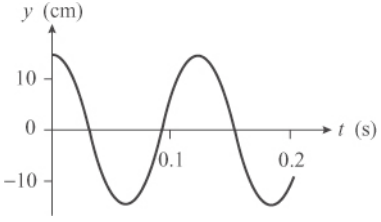
$$\boxed{y = 0.100\sin(\pi t)}$$

(f) $$\boxed{y = 0.100\sin(4.71 - \pi t)}$$

(g) $$v_y = \frac{\partial y}{\partial t} = 0.100\text{ m}(-3.14/\text{s})\cos(3.14x/\text{m} - 3.14t/\text{s})$$

The cosine varies between +1 and −1, so maximum $v_y = \boxed{0.314\text{ m/s}}$.

P16.14 (a) ANS. FIG. P16.14 shows the y vs. t plot of the given wave.



ANS. FIG. P16.14

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(b) The time from one peak to the next one is

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{50.3\ \text{s}^{-1}} = \boxed{0.125\ \text{s}}$$

(c) This agrees with the period found in the example in the text.

P16.15 The wave function is given as

$$y = (0.120\ \text{m}) \sin\left(\frac{\pi}{8}x + 4\pi t\right)$$

(a) We differentiate the wave function with respect to time to obtain the velocity:

$$v = \frac{\partial y}{\partial t}: \quad v = (0.120)(4\pi)\cos\left(\frac{\pi}{8}x + 4\pi t\right)$$

$$v(0.200\ \text{s}, 1.60\ \text{m}) = \boxed{-1.51\ \text{m/s}}$$

(b) Differentiating the velocity function gives the acceleration:

$$a = \frac{\partial v}{\partial t}: \quad a = (-0.120\ \text{m})(4\pi)^2 \sin\left(\frac{\pi}{8}x + 4\pi t\right)$$

$$a(0.200\ \text{s}, 1.60\ \text{m}) = \boxed{0}$$

(c) $k = \dfrac{\pi}{8} = \dfrac{2\pi}{\lambda}$: $\lambda = \boxed{16.0\ \text{m}}$

(d) $\omega = 4\pi = \dfrac{2\pi}{T}$: $T = \boxed{0.500\ \text{s}}$

(e) $v = \dfrac{\lambda}{T} = \dfrac{16.0\ \text{m}}{0.500\ \text{s}} = \boxed{32.0\ \text{m/s}}$

P16.16 (a) At $x = 2.00$ m, $y = \boxed{0.100\sin(1.00 - 20.0t)}$. Because this disturbance varies sinusoidally in time, it describes simple harmonic motion.

(b) At $x = 2.00$ m, compare $y = 0.100\sin(1.00 - 20.0t)$ to $A\cos(\omega t + \phi)$:

$$\begin{aligned} y &= 0.100\sin(1.00 - 20.0t) = -0.100\sin(20.0t - 1.00) \\ &= 0.100\cos(20.0t - 1.00 + \pi) \\ &= 0.100\cos(20.0t + 2.14) \end{aligned}$$

so $\omega = 20.0$ rad/s and $f = \dfrac{\omega}{2\pi} = \boxed{3.18\ \text{Hz}}$

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P16.17 The wave function is: $y = 0.25 \sin (0.30x - 40t)$ m

Compare this with the general expression $y = A \sin (kx - \omega t)$:

(a) $A = \boxed{0.250 \text{ m}}$

(b) $\omega = \boxed{40.0 \text{ rad/s}}$

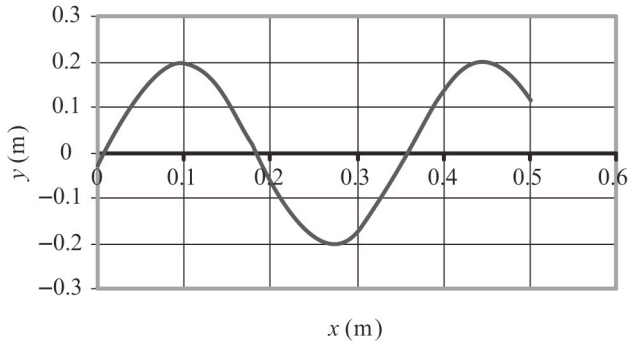
(c) $k = \boxed{0.300 \text{ rad/m}}$

(d) $\lambda = \dfrac{2\pi}{k} = \dfrac{2\pi}{0.300 \text{ rad/m}} = \boxed{20.9 \text{ m}}$

(e) $v = f\lambda = \left(\dfrac{\omega}{2\pi}\right)\lambda = \left(\dfrac{40.0 \text{ rad/s}}{2\pi}\right)(20.9 \text{ m}) = \boxed{133 \text{ m/s}}$

(f) The wave moves to the right, $\boxed{\text{in the} + x \text{ direction}}$.

P16.18 (a) ANS. FIG. P16.18(a) shows a sketch of the wave at $t = 0$.



ANS FIG. P16.18(a)

(b) $k = \dfrac{2\pi}{\lambda} = \dfrac{2\pi}{0.350 \text{ m}} = \boxed{18.0 \text{ rad/m}}$

(c) $T = \dfrac{1}{f} = \dfrac{1}{12.0/\text{s}} = \boxed{0.083\ 3 \text{ s}}$

(d) $\omega = 2\pi f = 2\pi 12.0/\text{s} = \boxed{75.4 \text{ rad/s}}$

(e) $|v| = f\lambda = (12.0/\text{s})(0.350 \text{ m}) = \boxed{4.20 \text{ m/s}}$

(f) $y = A \sin (kx + \omega t + \phi)$ specializes to

$$\boxed{y = (0.200 \text{ m}) \sin (18.0\, x/\text{m} + 75.4\, t/\text{s} + \phi)}$$

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(g) At $x = 0, t = 0$ we require

$$-3.00 \times 10^{-2}\ \text{m} = (0.200\ \text{m}) \sin(+\phi)$$
$$\phi = -8.63° = -0.151\ \text{rad}$$

so

$$\boxed{y(x,\ t) = 0.200 \sin(18.0x + 75.4t - 0.151),\ \text{where } x \text{ and } y \text{ are in meters and } t \text{ is in seconds.}}$$

P16.19 Using the traveling wave model, we can put constants with the right values into $y = A \sin(kx + \omega t + \phi)$ to have the mathematical representation of the wave. We have the same (positive) signs for both kx and ωt so that a point of constant phase will be at a decreasing value of x as t increases—that is, so that the wave will move to the left.

The amplitude is $A = y_{\max} = 8.00\ \text{cm} = 0.080\ 0\ \text{m}$

The wave number is $k = \dfrac{2\pi}{\lambda} = \dfrac{2\pi}{0.800\ \text{m}} = 2.50\pi\ \text{m}^{-1}$

The angular frequency is $\omega = 2\pi f = 2\pi(3.00\ \text{s}^{-1}) = 6.00\pi\ \text{rad/s}$

(a) In $y = A \sin(kx + \omega t + \phi)$, choosing $\phi = 0$ will make it true that $y(0, 0) = 0$. Then the wave function becomes upon substitution of the constant values for this wave

$$\boxed{y = (0.080\ 0) \sin(2.50\pi x + 6.00\pi t)}$$

(b) In general, $y = (0.080\ 0)\sin(2.50\pi x + 6.00\pi t + \phi)$

If $y(x, 0) = 0$ at $x = 0.100$ m, we require

$$0 = (0.080\ 0)\sin(2.50\pi + \phi)$$

so we must have the phase constant be $\phi = -0.250\pi$ rad.

Therefore, the wave function for all values of x and t is

$$\boxed{y = 0.080\ 0 \sin(2.50\pi x + 6.00\pi t - 0.250\pi),\ \text{where } x \text{ and } y \text{ are in meters and } t \text{ is in seconds.}}$$

P16.20 (a) Let us write the wave function as $y(x,t) = A\sin(kx + \omega t + \phi)$.

We have $y_i = y(0, 0) = A \sin\phi = 0.020\ 0\ \text{m}$

and $v_i = v(0, 0) = \left.\dfrac{\partial y}{\partial t}\right|_{0,\,0} = A\omega \cos\phi = -2.00\ \text{m/s}.$

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Also, $\omega = \dfrac{2\pi}{T} = \dfrac{2\pi}{0.025\ 0\ \text{s}} = 80.0\pi\ \text{s}^{-1}$.

Use the identity $\sin^2\phi + \cos^2\phi = 1$ and the expressions for y_i and v_i:

$$\frac{(A\sin\phi)^2}{A^2} + \frac{(A\omega\cos\phi)^2}{A^2\omega^2} = 1$$

$$(A\sin\phi)^2 + \frac{(A\omega\cos\phi)^2}{\omega^2} = A^2$$

$$A^2 = y_i^2 + \left(\frac{v_i}{\omega}\right)^2 = (0.020\ 0\ \text{m})^2 + \left(\frac{-2.00\ \text{m/s}}{80.0\pi\ \text{s}^{-1}}\right)^2$$

$$A = \boxed{0.021\ 5\ \text{m}}$$

(b) $$\frac{\omega y_i}{v_i} = \frac{\omega(A\sin\phi)}{\omega A\cos\phi} = \tan\phi \rightarrow \tan\phi = \frac{80.0\pi(0.020\ 0)}{-2.00} = -2.51$$

Your calculator's answer $\phi = \tan^{-1}(-2.51) = -1.19$ rad is an angle in the fourth quadrant with a negative sine and positive cosine, just the reverse of what is required. Recall on the unit circle, an angle with a negative tangent can be in either the second or fourth quadrant. The sine is positive and the cosine is negative in the second quadrant. The angle in the second quadrant is

$$\phi = \pi - 1.19\ \text{rad} = \boxed{1.95\ \text{rad}}$$

(c) $$v_{y,\ \max} = A\omega = (0.021\ 5\ \text{m})(80.0\pi/\text{s}) = \boxed{5.41\ \text{m/s}}$$

(d) $$\lambda = v_x T = (30.0\ \text{m/s})(0.025\ 0\ \text{s}) = 0.750\ \text{m}$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{0.750\ \text{m}} = 8.38\ \text{m}^{-1}, \qquad \omega = 80.0\pi\ \text{s}^{-1}$$

$$\boxed{y(x,\ t) = (0.021\ 5)\sin(8.38x + 80.0\pi t + 1.95)}$$

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Section 16.3 The Speed of Transverse Waves on Strings

P16.21 If the tension in the wire is T, the tensile stress is

$$\text{stress} = \frac{T}{A} \qquad \text{so} \qquad T = A(\text{stress})$$

The speed of transverse waves in the wire is

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{A(\text{Stress})}{m/L}} = \sqrt{\frac{\text{Stress}}{m/AL}} = \sqrt{\frac{\text{Stress}}{m/\text{Volume}}} = \sqrt{\frac{\text{Stress}}{\rho}}$$

where ρ is the density. The maximum velocity occurs when the stress is a maximum:

$$v_{\max} = \sqrt{\frac{2.70\times10^{8}\ \text{Pa}}{7860\ \text{kg/m}^3}} = \boxed{185\ \text{m/s}}$$

P16.22 The speed is given by

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{1\,350\ \text{kg}\cdot\text{m/s}^2}{5.00\times10^{-3}\ \text{kg/m}}} = \boxed{520\ \text{m/s}}$$

P16.23 The two wave speeds can be written as

$$v_1 = \sqrt{T_1/\mu} \qquad \text{and} \qquad v_2 = \sqrt{T_2/\mu}$$

Since μ is constant, $\mu = \frac{T_2}{v_2^2} = \frac{T_1}{v_1^2}$, and

$$T_2 = \left(\frac{v_2}{v_1}\right)^2 T_1 = \left(\frac{30.0\ \text{m/s}}{20.0\ \text{m/s}}\right)^2 (6.00\ \text{N}) = \boxed{13.5\ \text{N}}$$

P16.24 (a) For the first equation,

$$f = \frac{1}{T} \quad\rightarrow\quad T = \frac{1}{f} \quad\rightarrow\quad [T] = \frac{1}{[f]} = \frac{1}{\text{T}^{-1}} = \text{T}$$

$\boxed{\text{units are seconds}}$

$$v = \sqrt{\frac{T}{\mu}} \quad\rightarrow\quad T = \mu v^2 \rightarrow \quad [T] = \left[\mu v^2\right] = \frac{\text{M}}{\text{L}}\left(\frac{\text{L}}{\text{T}}\right)^2 = \frac{\text{ML}}{\text{T}^2}$$

$\boxed{\text{units are newtons}}$

(b) $\boxed{\text{The first } T \text{ is period of time; the second is force of tension.}}$

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P16.25 The down and back distance is 4.00 m + 4.00 m = 8.00 m.

The speed is then $v = \frac{d_{\text{total}}}{t} = \frac{4(8.00\text{ m})}{0.800\text{ s}} = 40.0\text{ m/s} = \sqrt{\frac{T}{\mu}}$.

Now, $\mu = \frac{0.200\text{ kg}}{4.00\text{ m}} = 5.00 \times 10^{-2}\text{ kg/m}$.

So $T = \mu v^2 = (5.00 \times 10^{-2}\text{ kg/m})(40.0\text{ m/s})^2 = \boxed{80.0\text{ N}}$.

P16.26 (a) To write the equation, we determine the angular frequency and wave number:

$$\omega = 2\pi f = 2\pi(500\text{ Hz}) = 3\ 140\text{ rad/s}$$

$$k = \frac{\omega}{v} = \frac{3\ 140}{196} = 16.0\text{ m}^{-1}$$

$\boxed{y = (2.00 \times 10^{-4})\sin(16.0x - 3\,140t),\text{ where } y \text{ and } x \text{ are in meters and } t \text{ is in seconds.}}$

(b) $v = 196\text{ m/s} = \sqrt{\frac{T}{4.10 \times 10^{-3}\text{ kg/m}}} \rightarrow T = \boxed{158\text{ N}}$

P16.27 The total time interval is the sum of the two time intervals.

In each wire

$$\Delta t = \frac{L}{v} = L\sqrt{\frac{\mu}{T}}$$

Let A represent the cross-sectional area of one wire. The mass of one wire can be written both as $m = \rho V = \rho AL$ and also as $m = \mu L$.

Then we have $\mu = \rho A = \frac{\pi \rho d^2}{4}$.

Thus, $\Delta t = L\left(\frac{\pi \rho d^2}{4T}\right)^{1/2}$

For copper,

$$\Delta t = (20.0\text{ m})\left[\frac{(\pi)(8\ 920\text{ kg/m}^3)(1.00 \times 10^{-3}\text{ m})^2}{(4)(150\text{ N})}\right]^{1/2} = 0.137\text{ s}$$

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For steel,

$$\Delta t = (30.0\text{ m})\left[\frac{(\pi)(7\ 860\text{ kg/m}^3)(1.00\times10^{-3}\text{ m})^2}{(4)(150\text{ N})}\right]^{1/2} = 0.192\text{ s}$$

The total time interval is $0.137 + 0.192 = \boxed{0.329\text{ s}}$

P16.28 The tension in the string is $T = mg$, where g is the acceleration of gravity on the Moon, about one-sixth that of Earth. From the data given, what is the acceleration of gravity on the Moon?

The wave speed is

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{Mg}{m/L}} = \sqrt{\frac{MgL}{m}} = \frac{L}{t} \rightarrow \frac{MgL}{m} = \frac{L^2}{t^2} \rightarrow g = \frac{mL}{Mt^2}$$

$$g = \frac{mL}{Mt^2} = \frac{(4.00\times10^{-3}\text{ kg})(1.60\text{ m})}{(3.00\text{ kg})(26.1\times10^{-3}\text{ s})^2} = 3.13\text{ m/s}^2$$

The calculated gravitational acceleration of the Moon is almost twice that of the accepted value.

P16.29 (a) The tension in the string is

$$F = mg = (3.00\text{ kg})(9.80\text{ m/s}^2) = 29.4\text{ N}$$

Then, from $v = \sqrt{\frac{F}{\mu}}$, the mass per unit length is

$$\mu = \frac{F}{v^2} = \frac{29.4\text{ N}}{(24.0\text{ m/s})^2} = \boxed{0.0510\text{ kg/m}}$$

(b) When $m = 2.00$ kg, the tension is

$$F = mg = (2.00\text{ kg})(9.80\text{ m/s}^2) = 19.6\text{ N}$$

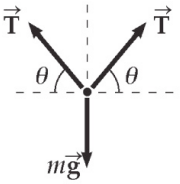
and the speed of transverse waves in the string is

$$v = \sqrt{\frac{F}{\mu}} = \sqrt{\frac{19.6\text{ N}}{0.0510\text{ kg/m}}} = \boxed{19.6\text{ m/s}}$$

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P16.30 From the free-body diagram $mg = 2T \sin\theta$

$$T = \frac{mg}{2\sin\theta}$$



ANS. FIG. P16.30

The angle θ is found from

$$\cos\theta = \frac{3L/8}{L/2} = \frac{3}{4}$$

$$\therefore \theta = 41.4°$$

(a) $$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{mg}{2\mu\sin\theta}}$$

$$= \sqrt{\frac{mg}{2\mu \sin 41.4°}} = \left(\sqrt{\frac{9.80\ \text{m/s}^2}{2\left(8.00\times10^{-3}\ \text{kg/m}\right)\sin 41.4°}}\right)\sqrt{m}$$

or $\boxed{v = (30.4\ )\sqrt{m}\text{, where } v \text{ is in meters per second and } m \text{ is in kilograms.}}$

(b) $v = 60.0 = 30.4\sqrt{m}$ and $\boxed{m = 3.89\ \text{kg}}$

P16.31 We use $v = \sqrt{\frac{T}{\mu}}$ to solve for the tension:

$$T = \mu v^2 = \rho A v^2 = \rho\pi r^2 v^2$$

$$T = \left(8920\ \text{kg/m}^3\right)(\pi)\left(7.50\times10^{-4}\ \text{m}\right)^2\left(200\ \text{m/s}\right)^2$$

$$T = \boxed{631\ \text{N}}$$

Section 16.5 Rate of Energy Transfer by Sinusoidal Waves on Strings

P16.32 (a) As for a string wave, the rate of energy transfer is proportional to the square of the amplitude and to the speed. The rate of energy transfer stays constant because each wavefront carries constant energy and the frequency stays constant. As the speed drops the amplitude must increase.

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(b) We write $P = FvA^2$, where F is some constant. With no absorption of energy,

$$Fv_{\text{granite}}A_{\text{granite}}^2 = Fv_{\text{mudfill}}A_{\text{mudfill}}^2$$

$$\frac{A_{\text{mudfill}}}{A_{\text{granite}}} = \sqrt{\frac{v_{\text{granite}}}{v_{\text{mudfill}}}} = \sqrt{\frac{v_{\text{granite}}}{v_{\text{granite}}/25.0}} = \sqrt{\frac{25.0v_{\text{granite}}}{v_{\text{granite}}}} = 5.00$$

The amplitude increases by 5.00 times.

P16.33 We are given T = constant; we use the equation for the speed of a wave on a string, $v = \sqrt{\frac{T}{\mu}}$, and the power supplied to a vibrating string, $P = \frac{1}{2}\mu\omega^2 A^2 v$.

(a) If L is doubled, μ is still the same, so v remains constant: therefore P is constant: 1.

(b) If A is doubled and ω is halved, $P \propto \omega^2 A^2$ remains constant: 1.

(c) If λ and A are doubled, the product $\omega^2 A^2 \propto \frac{A^2}{\lambda^2}$ remains constant, so 1.

(d) If L and λ are halved, μ is still the same, and $\omega^2 \propto \frac{1}{\lambda^2}$ is quadrupled, so P is increased by a factor of 4.

P16.34 We will use the expression for power carried by a wave on a string. The wave speed is $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{100\text{ N}}{4.00\times10^{-2}\text{ kg/m}}} = 50.0\text{ m/s}$

From $P = \frac{1}{2}\mu\omega^2 A^2 v$, we have

$$\omega^2 = \frac{2P}{\mu A^2 v} = \frac{2(300\text{ N}\cdot\text{m/s})}{(4.00\times10^{-2}\text{ kg/m})(5.00\times10^{-2}\text{ m})^2(50.0\text{ m/s})}$$

Computing,

$$\omega = 346\text{ rad/s} \quad \text{and} \quad f = \frac{\omega}{2\pi} = \boxed{55.1\text{ Hz}}$$

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P16.35 Comparing the given wave function, $y = (0.15)\sin(0.80x - 50t)$, with the general wave function, $y = A\sin(kx - \omega t)$, we have $k = 0.80$ rad/m, $\omega = 50$ rad/s, and $A = 0.15$ m.

(a) $$v = f\lambda = \frac{\omega}{2\pi}\frac{2\pi}{k} = \frac{\omega}{k} = \frac{50.0}{0.800}\ \text{m/s} = \boxed{62.5\ \text{m/s}}$$

(b) $$\lambda = \frac{2\pi}{k} = \frac{2\pi}{0.800}\ \text{m} = \boxed{7.85\ \text{m}}$$

(c) $$f = \frac{50.0}{2\pi} = \boxed{7.96\ \text{Hz}}$$

(d) $$P = \frac{1}{2}\mu\omega^2 A^2 v = \frac{1}{2}\left(12.0\times10^{-3}\right)(50.0)^2(0.150)^2(62.5)\ \text{W} = \boxed{21.1\ \text{W}}$$

P16.36 The frequency and angular frequency of the wave are

$$f = \frac{v}{\lambda} = \frac{30.0\ \text{m/s}}{0.500\ \text{s}} = 60.0\ \text{Hz} \quad \text{and} \quad \omega = 2\pi f = 120\pi\ \text{rad/s}$$

The power that is required is then

$$P = \frac{1}{2}\mu\omega^2 A^2 v$$
$$= \frac{1}{2}\left(\frac{0.180\ \text{kg}}{3.60\ \text{m}}\right)(120\pi\ \text{rad/s})^2(0.100\ \text{m})^2(30.0\ \text{m/s})$$
$$= \boxed{1.07\ \text{kW}}$$

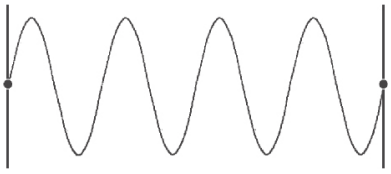
P16.37 We are given $\mu = 30.0\ \text{g/m} = 30.0\times10^{-3}\ \text{kg/m}$, with

$$\lambda = 1.50\ \text{m}$$
$$f = 50.0\ \text{Hz}: \quad \omega = 2\pi f = 314\ \text{s}^{-1}$$
$$2A = 0.150\ \text{m}: \quad A = 7.50\times10^{-2}\ \text{m}$$

(a) From $y = A\sin\left(\frac{2\pi}{\lambda}x - \omega t\right)$, $\boxed{y = (0.075)\sin(4.19x - 314t)}$

(b) $$P = \frac{1}{2}\mu\omega^2 A^2 v = \frac{1}{2}\left(30.0\times10^{-3}\right)(314)^2\left(7.50\times10^{-2}\right)^2\left(\frac{314}{4.19}\right)\text{W}$$

$\boxed{P = 625\ \text{W}}$



ANS. FIG. P16.37

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P16.38 Originally,

$$P_0 = \frac{1}{2}\mu\omega^2 A^2 v$$

$$P_0 = \frac{1}{2}\mu\omega^2 A^2 \sqrt{\frac{T}{\mu}}$$

$$P_0 = \frac{1}{2}\omega^2 A^2 \sqrt{T\mu}$$

The doubled string will have doubled mass per length. Presuming that we hold tension constant, it can carry power larger by $\sqrt{2}$ times:

$$P = \frac{1}{2}\omega^2 A^2 \sqrt{T(2\mu)} = \sqrt{2}\left(\frac{1}{2}\omega^2 A^2 \sqrt{T\mu}\right) = \boxed{\sqrt{2}P_0}$$

P16.39 Comparing

$$y = 0.350\sin\left(10\pi t - 3\pi x + \frac{\pi}{4}\right)$$

with

$$y = A\sin\left(kx - \omega t + \phi\right) = A\sin\left(\omega t - kx - \phi + \pi\right)$$

we have

$$k = 3\pi \text{ m}^{-1},\ \omega = 10\pi \text{ s}^{-1},\ \text{and } A = 0.350 \text{ m}$$

Then,

$$v = f\lambda = \left(2\pi f\right)\left(\frac{\lambda}{2\pi}\right) = \frac{\omega}{k} = \frac{10\pi \text{ s}^{-1}}{3\pi \text{ m}^{-1}} = 3.33 \text{ m/s}$$

(a) The rate of energy transport is

$$P = \frac{1}{2}\mu\omega^2 A^2 v$$

$$= \frac{1}{2}\left(75\times10^{-3} \text{ kg/m}\right)\left(10\pi \text{ s}^{-1}\right)^2 (0.350 \text{ m})^2 (3.33 \text{ m/s})$$

$$= \boxed{15.1 \text{ W}}$$

(b) Recall that $vT = \lambda$. The energy per cycle is

$$E_\lambda = P\,T = \frac{1}{2}\mu\omega^2 A^2 \lambda$$

$$= \frac{1}{2}\left(75.0\times10^{-3} \text{ kg/m}\right)\left(10\pi \text{ s}^{-1}\right)^2 (0.350 \text{ m})^2 \left(\frac{2\pi}{3\pi \text{ m}^{-1}}\right)$$

$$= \boxed{3.02 \text{ J}}$$

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P16.40 Suppose that no energy is absorbed or carried down into the water. Then a fixed amount of power is spread thinner farther away from the source. It is spread over the circumference $2\pi r$ of an expanding circle. The power-per-width across the wave front

$$\frac{P}{2\pi r}$$

is proportional to amplitude squared, so amplitude is proportional to

$$\sqrt{\frac{P}{2\pi r}}$$

Section 16.6 The Linear Wave Equation

P16.41 The important thing to remember with partial derivatives is that **you treat all variables as constants, except the single variable of interest**. Keeping this in mind, we must apply two standard rules of differentiation to the function $y = \ln[b(x - vt)]$:

$$\frac{\partial}{\partial x}\left[\ln f(x)\right] = \frac{1}{f(x)}\frac{\partial\left[f(x)\right]}{\partial x} \qquad \textbf{[1]}$$

$$\frac{\partial}{\partial x}\left[\frac{1}{f(x)}\right] = \frac{\partial}{\partial x}\left[f(x)\right]^{-1} = (-1)\left[f(x)\right]^{-2}\frac{\partial\left[f(x)\right]}{\partial x}$$

$$= -\frac{1}{\left[f(x)\right]^{2}}\frac{\partial\left[f(x)\right]}{\partial x} \qquad \textbf{[2]}$$

Applying [1],

$$\frac{\partial y}{\partial x} = \left(\frac{1}{b(x - vt)}\right)\frac{\partial(bx - bvt)}{\partial x} = \left(\frac{1}{b(x - vt)}\right)(b) = \frac{1}{x - vt}$$

Applying [2],

$$\frac{\partial^2 y}{\partial x^2} = -\frac{1}{(x - vt)^2}$$

In a similar way,

$$\frac{\partial y}{\partial t} = \frac{-v}{x - vt} \quad \text{and} \quad \frac{\partial^2 y}{\partial t^2} = \frac{v^2}{(x - vt)^2}$$

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From the second-order partial derivatives, we see that it is true that

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$$

so the proposed function **is** one solution to the wave equation.

P16.42 (a) $A = (7.00 + 3.00)(4.00)$ yields $\boxed{A = 40.0}$

(b) $\boxed{A = 7.00,\ B = 0,\ \text{and}\ C = 3.00}$

(c) In order for two vectors to be equal, they must have the same magnitude and the same direction in three-dimensional space. All of their components must be equal, so all coefficients of the unit vectors must be equal.

(d) $\boxed{A = 0}$ $\boxed{B = 7.00}$ in meters, $\boxed{C = 3.00}$ in m^{-1}, $\boxed{D = 4.00}$ in s^{-1}, $\boxed{E = 2.00}$ in rad.

(e) Identify corresponding parts. In order for two functions to be identically equal, corresponding parts must be identical. The argument of the sine function must have no units, or be equivalent to units of radians.

P16.43 The linear wave equation is $\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$.

If $y = e^{b(x-vt)}$

Then $\frac{\partial y}{\partial t} = -bve^{b(x-vt)}$ and $\frac{\partial y}{\partial x} = be^{b(x-vt)}$

$\frac{\partial^2 y}{\partial t^2} = b^2v^2e^{b(x-vt)}$ and $\frac{\partial^2 y}{\partial x^2} = b^2e^{b(x-vt)}$

Therefore, $\frac{\partial^2 y}{\partial t^2} = v^2\frac{\partial^2 y}{\partial x^2}$, demonstrating that $e^{b(x-vt)}$ is a solution.

P16.44 (a) From $y = x^2 + v^2t^2$,

evaluate $\frac{\partial y}{\partial x} = 2x$ and $\frac{\partial^2 y}{\partial x^2} = 2$

Also, $\frac{\partial y}{\partial t} = v^2 2t$ and $\frac{\partial^2 y}{\partial t^2} = 2v^2$

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Does $\frac{\partial^2 y}{\partial t^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$?

By substitution, we must test $2 = \frac{1}{v^2}\left(2v^2\right)$ and this is true, so the wave function does satisfy the wave equation.

(b) Note

$$\frac{1}{2}(x+vt)^2 + \frac{1}{2}(x-vt)^2 = \frac{1}{2}x^2 + xvt + \frac{1}{2}v^2t^2 + \frac{1}{2}x^2 - xvt + \frac{1}{2}v^2t^2$$
$$= x^2 + v^2t^2$$

as required. So

$$\boxed{f(x+vt) = \frac{1}{2}(x+vt)^2} \quad \text{and} \quad \boxed{g(x-vt) = \frac{1}{2}(x-vt)^2}$$

(c) $y = \sin x \cos vt$ makes

$$\frac{\partial y}{\partial x} = \cos x \cos vt \qquad \frac{\partial^2 y}{\partial x^2} = -\sin x \cos vt$$

$$\frac{\partial y}{\partial t} = -v \sin x \sin vt \qquad \frac{\partial^2 y}{\partial t^2} = -v^2 \sin x \cos vt$$

Then $\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$ becomes $-\sin x \cos vt = \frac{-1}{v^2}v^2 \sin x \cos vt$ which is true, as required.

Note $\sin(x+vt) = \sin x \cos vt + \cos x \sin vt$

$\sin(x-vt) = \sin x \cos vt - \cos x \sin vt$

So $\sin x \cos vt = f(x+vt) + g(x-vt)$ with

$$\boxed{f(x+vt) = \frac{1}{2}\sin(x+vt)} \quad \text{and} \quad \boxed{g(x-vt) = \frac{1}{2}\sin(x-vt)}$$

Additional Problems

P16.45 The equation $v = \lambda f$ is a special case of

speed = (cycle length)(repetition rate)

Thus,

$$v = \left(19.0 \times 10^{-3} \text{ m/frame}\right)\left(24.0 \text{ frames/s}\right) = \boxed{0.456 \text{ m/s}}$$

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P16.46 Assume a typical distance between adjacent people ~ 1 m.

Then the wave speed is $v = \frac{\Delta x}{\Delta t} \sim \frac{1\text{ m}}{0.1\text{ s}} \sim 10\text{ m/s}$.

Model the stadium as a circle with a radius of order 100 m. Then, the time for one circuit around the stadium is

$$T = \frac{2\pi r}{v} \sim \frac{2\pi\left(10^2\right)}{10\text{ m/s}} = 63\text{ s} \boxed{\sim 1\text{ min}}$$

P16.47 The speed of the wave on the rope is $v = \sqrt{\frac{T}{\mu}}$ and in this case $T = mg$;

therefore, $m = \frac{\mu v^2}{g}$.

Now $v = f\lambda$ implies $v = \frac{\omega}{k}$ so that

$$m = \frac{\mu}{g}\left(\frac{\omega}{k}\right)^2 = \frac{0.250\text{ kg/m}}{9.80\text{ m/s}^2}\left[\frac{18\pi\text{ s}^{-1}}{0.750\pi\text{ m}^{-1}}\right]^2 = \boxed{14.7\text{ kg}}$$

***P16.48** $v = \frac{2d}{t}$ gives

$$d = \frac{vt}{2} = \frac{1}{2}\left(6.50\times10^3\text{ m/s}\right)(1.85\text{ s}) = \boxed{6.01\text{ km}}$$

P16.49 The block-cord-Earth system is isolated, so energy is conserved as the block moves down distance x:

$$\Delta K + \Delta U = 0 \rightarrow$$

$$\left(K + U_g + U_s\right)_{\text{top}} = \left(K + U_g + U_s\right)_{\text{bottom}}$$

$$0 + Mgx + 0 + 0 = 0 + 0 + \frac{1}{2}kx^2$$

$$x = \frac{2Mg}{k}$$

(a) $T = kx = 2Mg = 2(2.00\text{ kg})\left(9.80\text{ m/s}^2\right) = \boxed{39.2\text{ N}}$

(b) $L = L_0 + x = L_0 + \frac{2Mg}{k}$

$$L = 0.500\text{ m} + \frac{39.2\text{ N}}{100\text{ N/m}} = \boxed{0.892\text{ m}}$$

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(c) $$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{TL}{m}}$$

$$v = \sqrt{\frac{39.2\ \text{N} \times 0.892\ \text{m}}{5.0 \times 10^{-3}\ \text{kg}}}$$

$$v = \boxed{83.6\ \text{m/s}}$$

P16.50 The block-cord-Earth system is isolated, so energy is conserved as the block moves down distance x:

$$\Delta K + \Delta U = 0 \rightarrow$$

$$\left(K + U_g + U_s\right)_{\text{top}} = \left(K + U_g + U_s\right)_{\text{bottom}}$$

$$0 + Mgx + 0 + 0 = 0 + 0 + \frac{1}{2}kx^2$$

$$Mgx = \frac{1}{2}kx^2$$

(a) $$T = kx = \boxed{2Mg}$$

(b) $$L = L_0 + x = \boxed{L_0 + \frac{2Mg}{k}}$$

(c) $$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{TL}{m}} = \boxed{\sqrt{\frac{2Mg}{m}\left(L_0 + \frac{2Mg}{k}\right)}}$$

P16.51 (a) The wave function becomes

$$0.175\ \text{m} = (0.350\ \text{m})\sin\left[(99.6\ \text{rad/s})t\right]$$

or $$\sin\left[(99.6\ \text{rad/s})t\right] = 0.500$$

The smallest two angles for which the sine function is 0.500 are 30.0° and 150°, i.e., 0.523 6 rad and 2.618 rad.

$$(99.6\ \text{rad/s})t_1 = 0.523\ 6\ \text{rad},\ \text{thus}\ t_1 = 5.26\ \text{ms}$$

$$(99.6\ \text{rad/s})t_2 = 2.618\ \text{rad},\ \text{thus}\ t_2 = 26.3\ \text{ms}$$

$$\Delta t \equiv t_2 - t_1 = 26.3\ \text{ms} - 5.26\ \text{ms} = \boxed{21.0\ \text{ms}}$$

(b) Distance traveled by the wave

$$= \left(\frac{\omega}{k}\right)\Delta t = \left(\frac{99.6\ \text{rad/s}}{1.25\ \text{rad/m}}\right)\left(21.0 \times 10^{-3}\ \text{s}\right) = \boxed{1.68\ \text{m}}$$

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P16.52 (a) From $y = (0.150\text{ m})\sin(0.800x - 50.0t) = A\sin(kx - \omega t)$

we compute

$$\partial y/\partial t = (0.150\text{ m})(-50.0\text{ s}^{-1})\cos(0.800x - 50.0t)$$

and $$a = \partial^2 y/\partial t^2 = -(0.150\text{ m})(-50.0\text{ s}^{-1})^2\sin(0.800x - 50.0t)$$

Then $a_{\max} = (0.150\text{ m})(50.0\text{ s}^{-1})^2 = \boxed{375\text{ m/s}^2}$

(b) For the 1.00-cm segment with maximum force acting on it,

$$\sum F = ma = \left(\frac{12.0\times10^{-3}\text{ kg}}{100\text{ cm}}\right)(1.00\text{ cm})(375\text{ m/s}^2) = \boxed{0.045\ 0\text{ N}}$$

(c) To find the tension in the string, we first compute the wave speed

$$v = \lambda f = \frac{\omega}{k} = \frac{50.0\text{ s}^{-1}}{0.800\text{ m}^{-1}} = 62.5\text{ m/s}$$

then,

$$v = \sqrt{\frac{T}{\mu}}\text{ gives } T = \mu v^2 = \left(\frac{12.0\times10^{-3}\text{ kg}}{1.00\text{ m}}\right)(62.5\text{ m/s})^2 = \boxed{46.9\text{ N}}$$

The maximum transverse force is very small compared to the tension, more than a thousand times smaller.

P16.53 Assuming the incline to be frictionless and taking the positive x direction to be up the incline:

$$\sum F_x = T - Mg\sin\theta = 0$$

or the tension in the string is $T = Mg\sin\theta$.

The speed of transverse waves in the string is then

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{Mg\sin\theta}{m/L}} = \sqrt{\frac{MgL\sin\theta}{m}}$$

The time interval for a pulse to travel the string's length is

$$\Delta t = \frac{L}{v} = L\sqrt{\frac{m}{MgL\sin\theta}} = \boxed{\sqrt{\frac{mL}{Mg\sin\theta}}}$$

P16.54 (a) The energy a wave crest carries is constant in the absence of absorption. Then the rate at which energy passes a stationary point, which is the power of the wave, is constant.

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(b) The power is proportional to the square of the amplitude and to the wave speed. The speed decreases as the wave moves into shallower water near shore, so the amplitude must increase.

(c) For the wave described, with a single direction of energy transport, the power is the same at the deep-water location ① and at the place ② with depth 9.00 m. Because power is proportional to the square of the amplitude and the wave speed, to express the constant power we write,

$$A_1^2 v_1 = A_2^2 v_2 = A_2^2 \sqrt{g d_2}$$

$$(1.80\text{ m})^2 (200\text{ m/s}) = A_2^2 \sqrt{(9.80\text{ m/s}^2)(9.00\text{ m})}$$

$$= A_2^2 (9.39\text{ m/s})$$

$$A_2 = 1.80\text{ m} \left(\frac{200\text{ m/s}}{9.39\text{ m/s}} \right)^{1/2}$$

$$= \boxed{8.31\text{ m}}$$

(d) As the water depth goes to zero, our model would predict zero speed and infinite amplitude. In fact the amplitude must be finite as the wave comes ashore. As the speed decreases the wavelength also decreases. When it becomes comparable to the water depth, or smaller, our formula $\sqrt{gd}$ for wave speed no longer applies.

P16.55 Let M = mass of block, m = mass of string. For the block, $\sum F = ma$ implies $T = \frac{mv_b^2}{r} = m\omega^2 r$. The speed of a wave on the string is then

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{M\omega^2 r}{m/r}} = r\omega\sqrt{\frac{M}{m}}$$

the travel time of the wave on the string is given by

$$\Delta t = \frac{r}{v} = \frac{1}{\omega}\sqrt{\frac{m}{M}}$$

and the angle through which the block rotates is

$$\Delta\theta = \omega\,\Delta t = \sqrt{\frac{m}{M}} = \sqrt{\frac{0.003\ 2\text{ kg}}{0.450\text{ kg}}} = \boxed{0.084\ 3\text{ rad}}$$

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P16.56 The transverse wave velocity in the string is $v_{\text{trans}} = \sqrt{\frac{T}{\mu}}$,

where T is the tension in the cord, and μ is the mass per unit length of the cord. The tension T is generated by the centripetal force holding the mass and cord in uniform circular motion at the angular velocity ω; thus:

$$T = F_c = M\frac{v^2}{r} = M\omega^2 r$$

where we note that M is the mass of the block.

The mass density of the cord is $\mu = \frac{m}{r}$; thus, the transverse wave velocity is

$$v_{\text{trans}} = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{\left(M\omega^2 r\right)}{\left(\frac{m}{r}\right)}} = \sqrt{\frac{\left(M\omega^2 r^2\right)}{\left(m\right)}} = \omega r\sqrt{\frac{M}{m}}$$

Now the transverse wave travels a distance r (the length of the cord) at a uniform velocity v_{trans}; thus, distance = $r = v_{\text{trans}}t$, and therefore,

$$t = \frac{r}{v_{\text{trans}}} = \frac{r}{\left(\omega r\sqrt{\frac{M}{m}}\right)} = \boxed{\frac{1}{\omega}\sqrt{\frac{m}{M}}}$$

which we may solve numerically:

$$t = \frac{1}{\omega}\sqrt{\frac{m}{M}} = \frac{1}{(10.0\text{ rad/s})}\sqrt{\frac{0.003\ 20\text{ kg}}{0.450\text{ kg}}} = \boxed{8.43\times10^{-3}\text{ s}}$$

[See Note to P16.57.]

P16.57 The transverse wave velocity in the string is $v_{\text{trans}} = \sqrt{\frac{T}{\mu}}$,

where T is the tension in the cord, and μ is the mass per unit length of the cord. The tension T is generated by the centripetal force holding the mass and cord in uniform circular motion at the angular velocity, ω; thus

$$T = F_c = M\frac{v^2}{r} = M\omega^2 r$$

where we note that M is the mass of the block, and the mass density of

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the cord is $\mu = \frac{m}{r}$. Thus transverse wave velocity is

$$v_{\text{trans}} = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{\left(M\omega^2 r\right)}{\left(\frac{m}{r}\right)}} = \sqrt{\frac{\left(M\omega^2 r^2\right)}{\left(m\right)}} = \omega r\sqrt{\frac{M}{m}}$$

Now the transverse wave travels a distance r (the length of the cord) at a uniform velocity v_{trans}; thus, distance = $r = v_{\text{trans}}t$, and therefore,

$$t = \frac{r}{v_{\text{trans}}} = \frac{r}{\left(\omega r\sqrt{\frac{M}{m}}\right)} = \boxed{\frac{1}{\omega}\sqrt{\frac{m}{M}}}$$

[Note: To solve this problem without integration of the mass density μ over the length of the cord to include the cord's own mass as a contribution to its own tension, and thus to a ***nonuniform tension*** along the length of the cord (and thus also to a ***nonuniform wave velocity*** along the cord), we must assume that the mass of the cord m is very small compared to the mass of the block M. In such a case, the mass of the cord does not contribute to the centripetal force, or as a result, to the tension on the cord itself. The only role the cord's mass will then play is in generating the linear density in the transverse wave velocity equation. To be forced to include mass of the cord in the centripetal force calculation is a ***significantly*** more difficult problem and is not attempted here.]

P16.58 (a) In $P = \frac{1}{2}\mu\omega^2 A^2 v$ where v is the wave speed, the quantity ωA is the maximum particle speed $v_{y,\max}$. We have $\mu = 0.500 \times 10^{-3}$ kg/m and

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{20.0\text{ N}}{0.500\times10^{-3}\text{ kg/m}}} = 200\text{ m/s}$$

Then

$$P = \frac{1}{2}\left(0.500\times10^{-3}\text{ kg/m}\right)v_{y,\max}^2\left(200\text{ m/s}\right)$$

$\boxed{P = 0.050\ 0\ v_{y,\max}^2\text{, where } P \text{ is in watts and } v_{y,\max} \text{ is in meters per second}}$

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(b) The power is proportional to the square of the maximum particle speed.

(c) In time $t = (3.00\text{ m})/v = (3.00\text{ m})/(200\text{ m/s}) = 1.50 \times 10^{-2}$ s, all the energy in a 3.00-m length of string goes past a point. Therefore, the amount of this energy is

$$E = Pt = \left(0.050\ 0\text{ kg/s}\right)v_{y,\max}^2\ (0.015\text{ s}) = \left(7.50\times10^{-4}\text{ kg}\ \right)v_{y,\max}^2$$

The mass of this section is

$$m_{3.00-m} = \left(0.500\times10^{-3}\text{ kg/m}\right)(3.00\text{ m}) = 1.50\times10^{-3}\text{ kg}$$

so $\dfrac{1}{2}m_{3.00-m} = 7.50\times10^{-4}\text{ kg}$

$E = (7.5\times10^{-4})\ v_{y,\max}^2$, where E is in joules and $v_{y,\max}$ is in meters per second.

(d) $\boxed{\dfrac{1}{2}mv_{y,\max}^2}$

(e) $E = Pt = (0.050\ 0\text{ kg/s})\ v_{y,\max}^2\ (6.00\text{ s})$

$\rightarrow$ $E\ =\ 0.300\ v_{y,\max}^2$ where E is in joules and $v_{y,\max}$ is in meters per second.

P16.59 (a) $\mu = \dfrac{dm}{dL} = \rho A\dfrac{dx}{dx} = \rho A$

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T}{\rho A}} = \sqrt{\frac{T}{\left[\rho\left(ax+b\right)\right]}} = \sqrt{\frac{T}{\left[\rho\left(10^{-3}x+10^{-2}\right)\text{cm}^2\right]}}$$

With all SI units,

$v = \sqrt{\dfrac{T}{\left[\rho\left(1.00\times10^{-5}x+1.00\times10^{-6}\right)\right]}}$ where x is in meter, T is in newtons, and v is in meters per second.

(b) $v(0) = \sqrt{\dfrac{24.0}{\left[\left(2700\right)\left(0+10^{-6}\right)\right]}} = \boxed{94.3\text{ m/s}}$

$$v(10.0\ m) = \sqrt{\frac{24.0}{\left[\left(2700\right)\left(10^{-4}+10^{-6}\right)\right]}} = \boxed{9.38\text{ m/s}}$$

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P16.60 Imagine a short transverse pulse traveling from the bottom to the top of the rope. When the pulse is at position x above the lower end of the rope, the wave speed of the pulse is given by $v = \sqrt{\frac{T}{\mu}}$, where $T = \mu x g$ is the tension required to support the weight of the rope below position x.

Therefore, $v = \sqrt{gx}$.

But $v = \frac{dx}{dt}$, so that $dt = \frac{dx}{\sqrt{gx}}$

and $t = \int_0^L \frac{dx}{\sqrt{gx}} = \frac{1}{\sqrt{g}} \frac{\sqrt{x}}{\frac{1}{2}}\Big|_0^L \approx \boxed{2\sqrt{\frac{L}{g}}}$

P16.61 (a) $P(x) = \frac{1}{2}\mu\omega^2 A^2 v = \frac{1}{2}\mu\omega^2 A_0^2 e^{-2bx}\left(\frac{\omega}{k}\right) = \boxed{\frac{\mu\omega^3}{2k} A_0^2 e^{-2bx}}$

(b) $P(0) = \boxed{\frac{\mu\omega^3}{2k} A_0^2}$

(c) $\frac{P(x)}{P(0)} = \boxed{e^{-2bx}}$

P16.62 $v = \left(\frac{4\ 450 \times 10^3 \text{ m}}{5.88 \text{ h}}\right)\left(\frac{1 \text{ h}}{3\ 600 \text{ s}}\right) = 210 \text{ m/s}$

$d_{\text{avg}} = \frac{v^2}{g} = \frac{(210 \text{ m/s})^2}{9.80 \text{ m/s}^2} = 4\ 500 \text{ m}$

The given speed corresponds to an ocean depth that is greater than the average ocean depth, about 4 280 m.

P16.63 Young's modulus for the wire may be written as $Y = \frac{T/A}{\Delta L/L}$, where T is the tension maintained in the wire and ΔL is the elongation produced by this tension. Also, the mass density of the wire may be expressed as

$$\rho = \frac{\mu}{A}$$

The speed of transverse waves in the wire is then

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T/A}{\mu/A}} = \sqrt{\frac{Y(\Delta L/L)}{\rho}}$$

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and the strain in the wire is $\frac{\Delta L}{L} = \frac{\rho v^2}{Y}$.

If the wire is made of aluminum and $v = 100$ m/s, the strain is

$$\frac{\Delta L}{L} = \frac{\left(2.70\times10^3 \text{ kg/m}^3\right)(100 \text{ m/s})^2}{7.00\times10^{10} \text{ N/m}^2} = \boxed{3.86\times10^{-4}}$$

Challenge Problems

P16.64 Refer to Problem 60. At distance x from the bottom, the tension is $T = \left(\frac{mxg}{L}\right) + Mg$, so the wave speed is:

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{TL}{m}} = \sqrt{xg + \left(\frac{MgL}{m}\right)} = \frac{dx}{dt} \rightarrow dt = \frac{dx}{\sqrt{xg + \left(\frac{MgL}{m}\right)}}$$

(a) Then

$$t = \int_0^t dt = \int_0^L \left[xg + \left(\frac{MgL}{m}\right)\right]^{-1/2} dx$$

gives
$$t = \frac{1}{g} \frac{\left[xg + \left(MgL/m\right)\right]^{1/2}}{\frac{1}{2}} \Bigg|_{x=0}^{x=L}$$

$$t = \frac{2}{g}\left[\left(Lg + \frac{MgL}{m}\right)^{1/2} - \left(\frac{MgL}{m}\right)^{1/2}\right]$$

$$\boxed{t = 2\sqrt{\frac{L}{mg}}\left(\sqrt{M+m} - \sqrt{M}\right)}$$

(b) When $M = 0$,

$$t = 2\sqrt{\frac{L}{g}}\left(\frac{\sqrt{m} - 0}{\sqrt{m}}\right) = \boxed{2\sqrt{\frac{L}{g}}}$$

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(c) As $m \to 0$ we expand

$$\sqrt{M+m} = \sqrt{M}\left(1+\frac{m}{M}\right)^{1/2} = \sqrt{M}\left(1+\frac{1}{2}\frac{m}{M}-\frac{1}{8}\frac{m^2}{M^2}+\cdots\right)$$

to obtain $$t = 2\sqrt{\frac{L}{mg}}\left(\sqrt{M}+\frac{1}{2}\left(\frac{m}{\sqrt{M}}\right)-\frac{1}{8}\left(m^2/M^{3/2}\right)+\cdots-\sqrt{M}\right)$$

$$t \approx 2\sqrt{\frac{L}{g}}\left(\frac{1}{2}\sqrt{\frac{m}{M}}\right) = \boxed{\sqrt{\frac{mL}{Mg}}}$$

where we neglect terms $\frac{1}{8}\left(\frac{m^2}{M^{3/2}}\right)$ and higher because terms with m^2 and higher powers are very small.

P16.65 (a) Refer to Problem 60. From the definition of velocity, find the relationship between the position x of the pulse and the time interval Δt required to reach that position from the bottom of the rope:

$$v = \frac{dx}{dt} \quad \rightarrow \quad dt = \frac{dx}{v} = \frac{dx}{\sqrt{gx}} \quad \rightarrow \quad \Delta t = \int \frac{dx}{\sqrt{gx}} \quad \rightarrow \quad \Delta t = 2\sqrt{\frac{x}{g}}$$

Evaluate this time interval for $x = \frac{L}{2}$:

$$\Delta t = 2\sqrt{\frac{L/2}{g}} = 2\sqrt{\frac{L}{2g}} = \frac{1}{\sqrt{2}}\left(2\sqrt{\frac{L}{g}}\right) = \boxed{0.707\left(2\sqrt{\frac{L}{g}}\right)}$$

(b) Solve the expression from part (a) for x and substitute the given time interval:

$$x = \frac{g(\Delta t)^2}{4} = \frac{g\left(\sqrt{L/2}\right)^2}{4} = \frac{g}{4}\frac{L}{g} = \boxed{\frac{L}{4}}$$

P16.66 (a) $\mu(x)$ is a linear function, so it is of the form $\mu(x) = mx + b$.

To have $\mu(0) = \mu_0$ we require $b = \mu_0$. Then $\mu(L) = \mu_L = mL + \mu_0$

so $m = \frac{\mu_L - \mu_0}{L}$.

Then $\boxed{\mu(x) = \frac{(\mu_L - \mu_0)x}{L} + \mu_0}$.

(b) Imagine the crest of a short transverse pulse traveling from one end of the string to the other. Consider the pulse to be at position

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x. From $v = \dfrac{dx}{dt}$, the time interval required to move from x to $x + dx$ is $\dfrac{dx}{v}$. The time interval required to move from 0 to L is

$$\Delta t = \int_0^L \frac{dx}{v} = \int_0^L \frac{dx}{\sqrt{T/\mu}} = \frac{1}{\sqrt{T}} \int_0^L \sqrt{\mu(x)}\, dx$$

$$\Delta t = \frac{1}{\sqrt{T}} \int_0^L \left(\frac{(\mu_L - \mu_0)x}{L} + \mu_0 \right)^{1/2} \left(\frac{\mu_L - \mu_0}{L} \right) dx \left(\frac{L}{\mu_L - \mu_0} \right)$$

$$\Delta t = \frac{1}{\sqrt{T}} \left(\frac{L}{\mu_L - \mu_0} \right) \left(\frac{(\mu_L - \mu_0)x}{L} + \mu_0 \right)^{3/2} \frac{1}{\left(\frac{3}{2}\right)} \Bigg|_0^L$$

$$\boxed{\Delta t = \frac{2L}{3\sqrt{T}(\mu_L - \mu_0)} \left(\mu_L^{3/2} - \mu_0^{3/2} \right)}$$

P16.67 (a) Consider a short section of chain at the top of the loop. A free-body diagram is shown. Its length is $s = R(2\Delta\theta)$ and its mass is $\mu R 2\Delta\theta$. In the frame of reference of the center of the loop, Newton's second law is



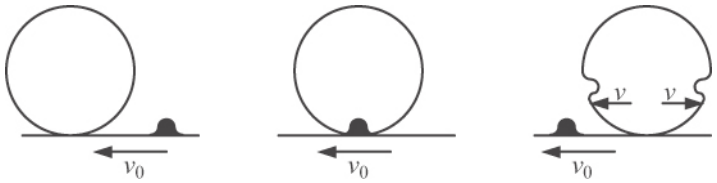
ANS. FIG. P16.67(a)

$$\sum F_y = ma_y: \quad 2T \sin\Delta\theta \text{ down} = \frac{mv_0^2}{R} \text{ down} = \frac{\mu R 2\Delta\theta v_0^2}{R}$$

For a very short section, $\sin\Delta\theta = \Delta\theta$ and $\boxed{T = \mu v_0^2}$

(b) The wave speed is $v = \sqrt{\dfrac{T}{\mu}} = \boxed{v_0}$

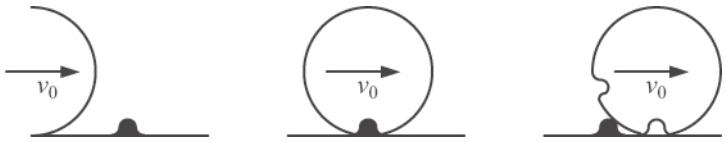
(c) In the frame of reference of the center of the loop, each pulse moves with equal speed clockwise and counterclockwise (ANS. FIG. P16.67(c1)).



ANS. FIG. P16.67(c1)

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In the frame of reference of the ground, once pulse moves backward, clockwise, at speed $v_0 + v = 2v_0$ and the other forward, counterclockwise, at $v_0 - v = 0$ (ANS. FIG. P16.67(c2))



ANS. FIG. P16.67(c2)

While the loop makes one revolution, the one pulse traveling clockwise makes two revolutions and the other pulse traveling counterclockwise does not move around the loop. The counter-clockwise pulse it is generated at the 6 o'clock position, and it will stay at the 6 o'clock position.

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P16.2 (a) See ANS. FIG. P16.2(a); (b) See ANS. FIG. P16.2(b); (c) The graph in ANS. FIG. P16.2(b) has the same amplitude and wavelength as the graph in ANS. FIG. P16.2(a). It differs just by being shifted toward larger x by 2.40 m; (d) The wave has traveled $d = vt = 2.40$ m to the right.

P16.4 (a) longitudinal P wave; (b) 666 s

P16.6 (a) See ANS. FIG. P16.6(a); (b) See ANS. FIG. P16.6(b); (c) See ANS. FIG. P16.6(c); (d) See ANS. FIG. P16.6(d); (e) See ANS. FIG. P16.6(e)

P16.8 0.800 m/s

P16.10 2.40 m/s

P16.12 ±6.67 cm

P16.14 (a) See ANS FIG P16.14; (b) 0.125 s; (c) This agrees with the period found in the example in the text.

P16.16 (a) 0.100 sin (1.002–20.0t); (b) 3.18 Hz

P16.18 (a) See ANS FIG P13.12(a); (b) 18.0 rad/m; (c) 0.083 3 s; (d) 75.4 rad/s; (e) 4.20 m/s; (f) $y = (0.200\ \text{m}) \sin\left(18.0x / \text{m} + 75.4t / s + \phi\right)$; (g) $y(x, t) =$ 0.200 sin (18.0x + 75.4t – 0.151), where x and y are in meters and t is in seconds.

P16.20 (a) 0.021 5 m; (b) 1.95 rad; (c) 5.41 m/s;
(d) $y(x,t) = (0.021\ 5) \sin\left(8.38x + 80.0\pi t + 1.95\right)$

P16.22 520 m/s

P16.24 (a) units are seconds and newtons; (b) The first T is period of time; the second is force of tension.

P16.26 (a) $y = (2.00 \times 10^{-4}) \sin (16.0x - 3\ 140t)$, where y and x are in meters and t is in seconds; (b) 158 N

P16.28 The calculated gravitational acceleration of the Moon is almost twice that of the accepted value.

P16.30 (a) $v = (30.4)\sqrt{m}$ where v is in meters per second and m is in kilograms; (b) $m = 3.89$ kg

P16.32 (a) As for a string wave, the rate of energy transfer is proportional to the square of the amplitude to the speed. The rate of energy transfer stays constant because each wavefront carries constant energy, and the frequency stays constant. As the speed drops, the amplitude must increase; (b) The amplitude increases by 5.00 times

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P16.34 55.1 Hz

P16.36 1.07 kW

P16.38 $\sqrt{2}P_0$

P16.40 See P16.40 for the full explanation.

P16.42 (a) $A = 40.0$; (b) $A = 7.00$, $B = 0$, and $C = 3.00$; (c) In order for two vectors to be equal, they must have the same magnitude and the same direction in three-directional space. All of their components must be equal, so all coefficients of the unit vectors must be equal; (d) $A = 0$, $B = 7.00$, $C = 3.00$, $D = 4.00$, $E = 2.00$; (e) Identify corresponding parts. In order for two functions to be identically equal, corresponding parts must be identical. The argument of the sine function must have no units or be equal to units of radians.

P16.44 (a) See P16.44(a) for full explanation; (b) $f(x+vt)=\frac{1}{2}(x+vt)^2$ and $g(x-vt)=\frac{1}{2}(x-vt)^2$; (c) $f(x+vt)=\frac{1}{2}\sin(x+vt)$ and $g(x-vt)=\frac{1}{2}\sin(x-vt)$

P16.46 ~1 min

P16.48 6.01 km

P16.50 (a) $2\,Mg$; (b) $L_0+\frac{2\,Mg}{k}$; (c) $\sqrt{\frac{2\,Mg}{k}\left(L_0+\frac{2\,Mg}{k}\right)}$

P16.52 (a) 375 m/s^2; (b) 0.045 0 N; (c) 46.9 N. The maximum transverse force is very small compared to the tension, more than a thousand times smaller.

P16.54 (a) The energy a wave crest carries is constant in the absence of absorption. Then the rate at which energy passes a stationary point, which is the power of the wave, is constant; (b) The power is proportional to the square of the amplitude and to the wave speed. The speed decreases as the wave moves into shallower water near shore, so the amplitude must increase; (c) 8.31 m; (d) As the water depth goes to zero, our model would predict zero speed and infinite amplitude. In fact, the amplitude must be finite as the wave comes ashore. As the speed decreases, the wavelength also decreases. When it becomes comparable to the water depth, or smaller, our formula $\sqrt{gd}$ for wave speed no longer applies.

P16.56 8.43×10^{-3} s

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P16.58 (a) $P = 0.050\ 0\ v_{y,\max}^2$ where P is in watts and $v_{y,\max}$ is in meters per second; (b) The power is proportional to the square of the maximum particle speed; (c) $E = \left(7.50 \times 10^{-4}\right) v_{y,\max}^2$ where E is in joules and $v_{y,\max}$ is in meters per second; (d) $\frac{1}{2} m v_{y,\max}^2$; (e) $E = 0.300 v_{y,\max}^2$ where E is in joules and $v_{y,\max}$ is in meters per second

P16.60 $2\sqrt{\frac{L}{g}}$

P16.62 The given speed corresponds to an ocean depth that is greater than the average ocean depth, about 4 280 m.

P16.64 (a) $t = 2\sqrt{\frac{L}{g}}\left(\sqrt{M+m} - \sqrt{M}\right)$; (b) $2\sqrt{\frac{L}{g}}$; (c) $\sqrt{\frac{mL}{Mg}}$

P16.66 (a) $\mu(x) = \frac{(\mu_L - \mu_0)x}{L} + \mu_0$; (b) $\Delta t = \frac{2L}{3\sqrt{T}\,(\mu_L - \mu_0)}\left(\mu_L^{3/2} - \mu_0^{3/2}\right)$

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